SURF AI & Large scale computing

Amsterdam UMC

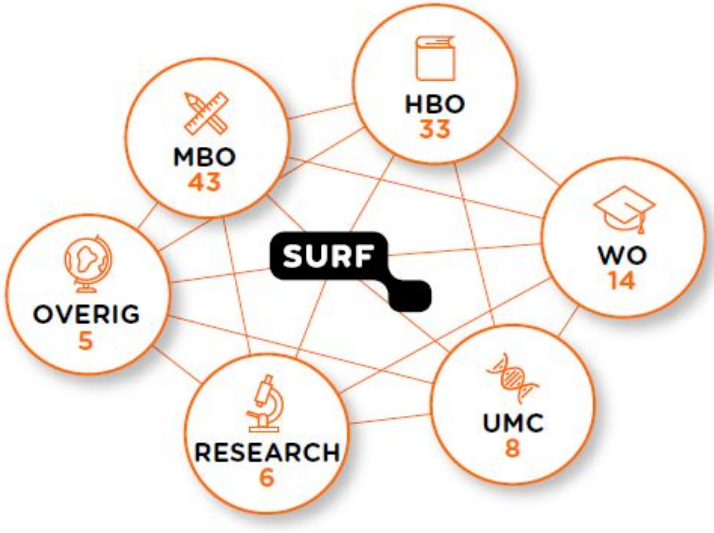
September 2025





Driving IT innovation in education and research

Together with our members, we ensure that the education and research community has access to the best and latest IT facilities for top-level research and talent development.



SURF Research services

Compute

We provide accessible and economical computing capabilities for researchers

High performance computing	Data Processing (HTC)	Cloud Computing
NL supercomputer Snellius	SPIDER	SURF Research Cloud
LUMI petascale supercomputer	GRID	HPC Cloud
Jules Verne exascale supercomputer		Public Clouds
NL AI Factory		SANE / Confidential Research Cloud
OSSC / Confidential computing		

Consultancy on HPC, HTC, AI / ML, Cloud & Visualization

We help both high demand, large scale users through **Snellius**, **Spider** and **GRID,** as well as provide accessible plug-and-play services through **Research Cloud** and **Public Cloud** providers.

Data

We help researchers organize, store, preserve and share data

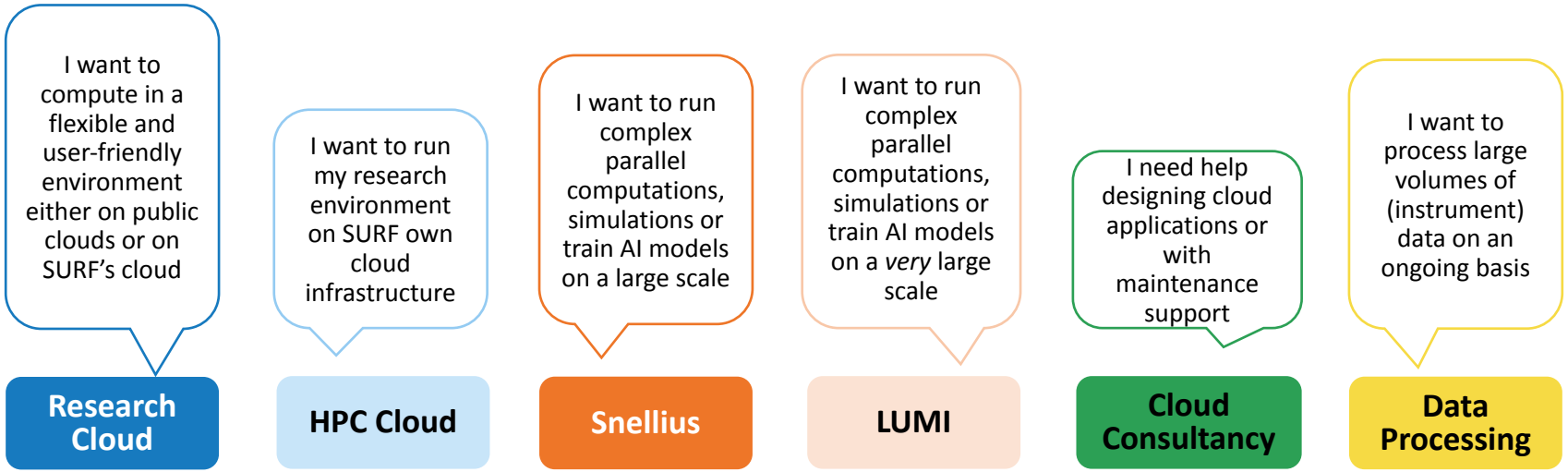
Sync & Share	General Storage	Long-term Storage	Research Data Management
Research Drive	Object Store	Data Archive	Yoda Hosting
SURF Drive	dCache		RDM Scale-out
SURF FileSender			iRODS hosting
			ICE

Consultancy on Storage, Preservation & Research Data Management

We provide trustworthy, secure & economical data storage for researchers through **Research-** & **SURF Drive**, **Object Store** and **Data Archive**. And promote data management according to the **FAIR principles**.

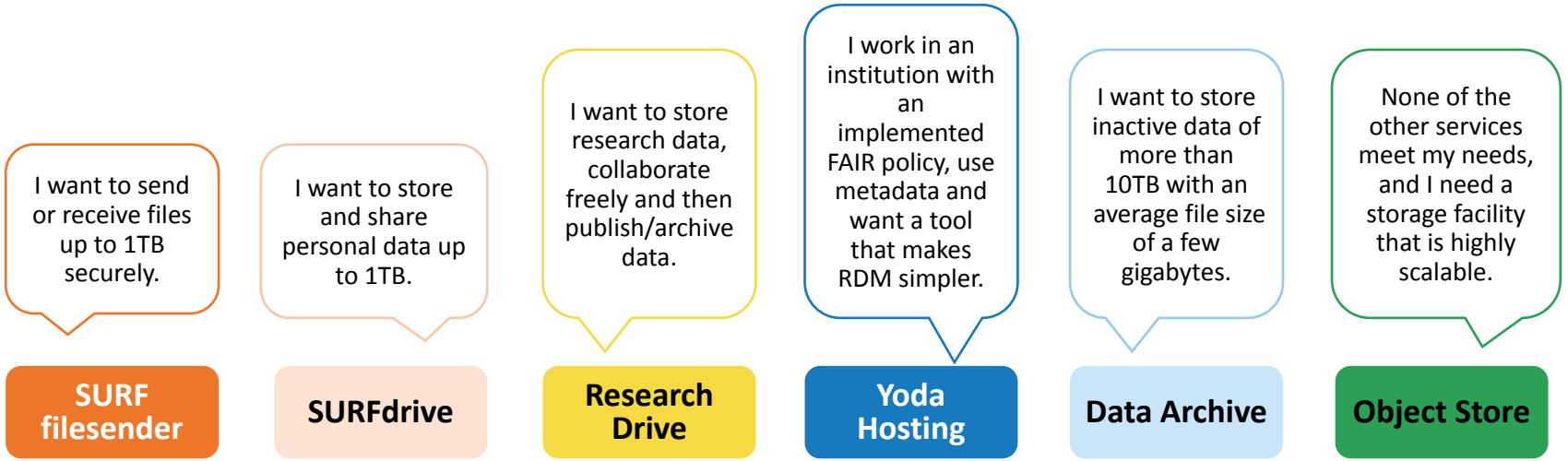
In production | **In development** | **Consultancy**

How do I compute with data?



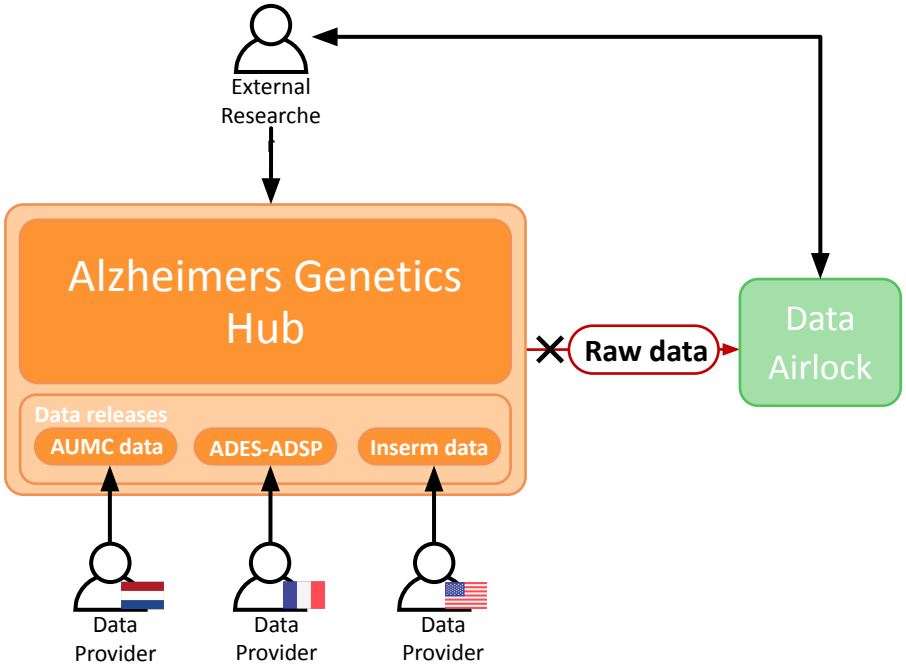
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Where do I store data?



Bespoke environments: The Alzheimer's Genetics Hub

- The AGH is a clone of Spider with significant extra security
- Data providers can only upload (not download!) data **into** the **storage** system
- The cluster is **disconnected** from the "normal" internet
- Files in and out go through an auditable "airlock"
- Airlock runs on Research Drive
- Storage in the airlock is not big enough for raw data



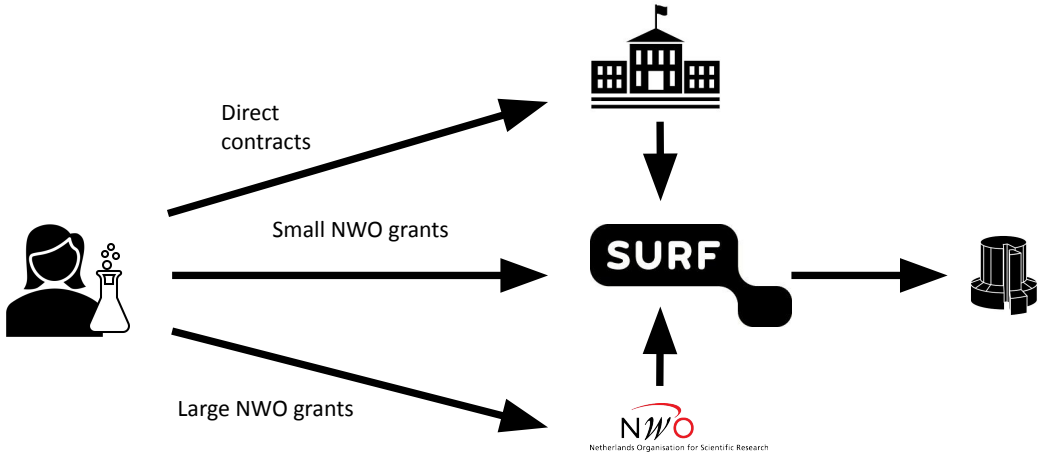
More info:
https://holstegelab.eu/launch-of-the-alzheimer-genetics-hub/

Working with sensitive data on our Compute services

Solution	Use case	Features	Suitable for sensitive data?
SURF Research Cloud	Small to medium scale computation, storage and exploration	▪ VM's on SURF private cloud ▪ VM's on public clouds ▪ User friendly interface ▪ Collaboration tools ▪ Shared application catalogue ▪ ISO27001	**Maybe** SRC can be used for sensitive data, if you take certain measures and precautions. Read more.
Data Processing	High-throughput computing	▪ Large scale high throughput capacity ▪ Data intensive workloads ▪ Customizable and dedicated environments ▪ Distributed processing (embarrassingly parallel) ▪ ISO27001	**Maybe** By default, we do not recommend using Data Processing for sensitive data. However, dedicated and customized 'trusted' clusters can be created to support sensitive data.
National Supercomputer	High performance computing	▪ High performance, large scale parallel computation ▪ Large variety of CPU, GPU and storage types. ▪ Extensive software library ▪ ISO27001	**Maybe** By default, we do not recommend using Snellius for sensitive data unless you take specific measures. Read more. For large, frequently used datasets, we can also establish permanent secure enclaves on the National Supercomputer (see below).
SANE on SURF Research Cloud	Small scale compute and exploration for sensitive data	▪ Data provider keeps control of data. ▪ Supports interactive and non-interactive analysis (blind). ▪ Leverage SRC catalogue items. ▪ Limited to one VM and limited access options.	**Yes**
OSSC on the National Supercomputer	Secure HPC enclave to work with CBS microdata	▪ High performance, large scale parallel computation ▪ Dedicated environment for CBS microdata. ▪ Access via CBSRA	**Yes**
OSCAR on the National Supercomputer	Secure HPC enclave dedicated to other frequently used sensitive datasets	▪ High performance, large scale parallel computation. ▪ Dedicated environment for particular dataset. ▪ Needs to be funded, configured and maintained long-term (5+ years). ▪ Only when there are 100s of potential applications or users.	**Yes**

Access routes to research services

- Direct contracts Give organisations direct access to SURF compute facilities on a contract basis.
- Small NWO grants To run a pilot project (1/year), you can apply for limited amounts of compute, storage and support directly via SURF.
- Large NWO grants for large scale* research projects need to apply via NWO.
- Open calls for public cloud access.



***Large Scale Defined:**

+1.00.000 CPU of GPU SBU **Snellius Supercomputer**
+50.000 CPU core-hours **HPC Cloud (Research Cloud)**
+5.000 GPU-hours **HPC Cloud (Research Cloud)**
+500.000 CPU core-hours **Grid/Spider**
+10.000 GPU-hours **Grid/Spider**
+100.000 CPU core-hours **Cloud Research Consultancy** (MS4/CCS)
+10.000 GPU hours **Cloud Research Consultancy** (MS4/CCS)

Incl. varying amounts of disk-, tape-storage, research drive and support.

SBU = System Billing Unit

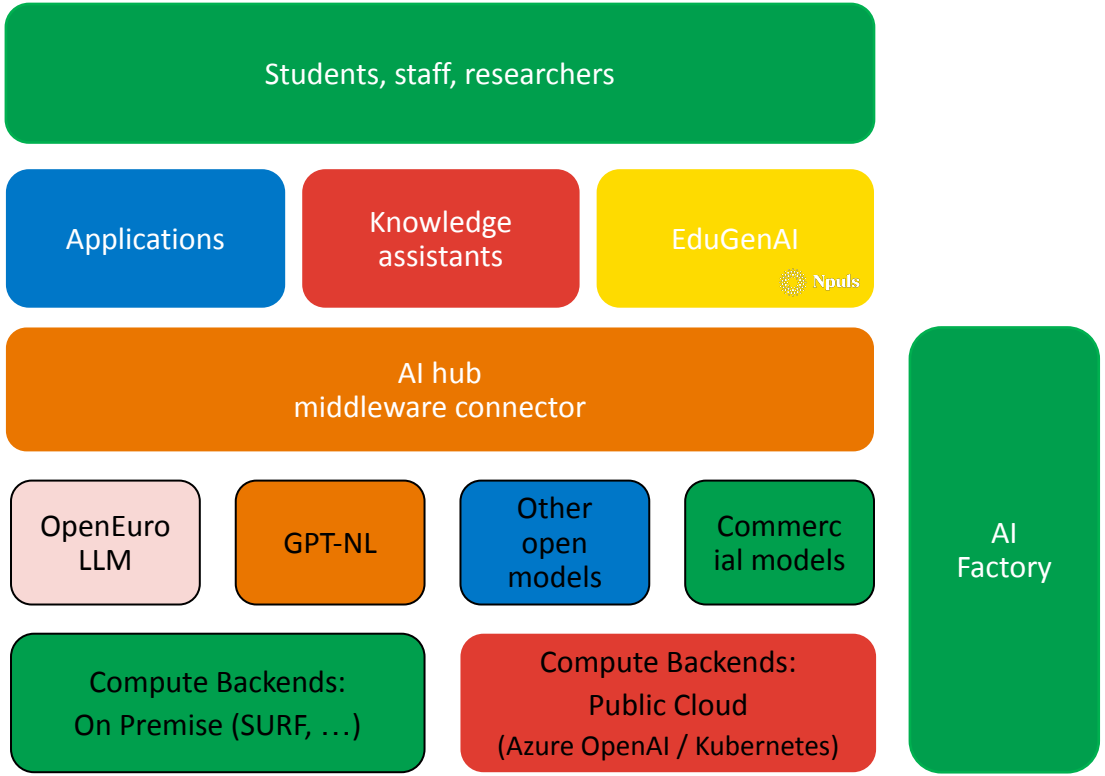
AI infrastructure that can be **trusted**



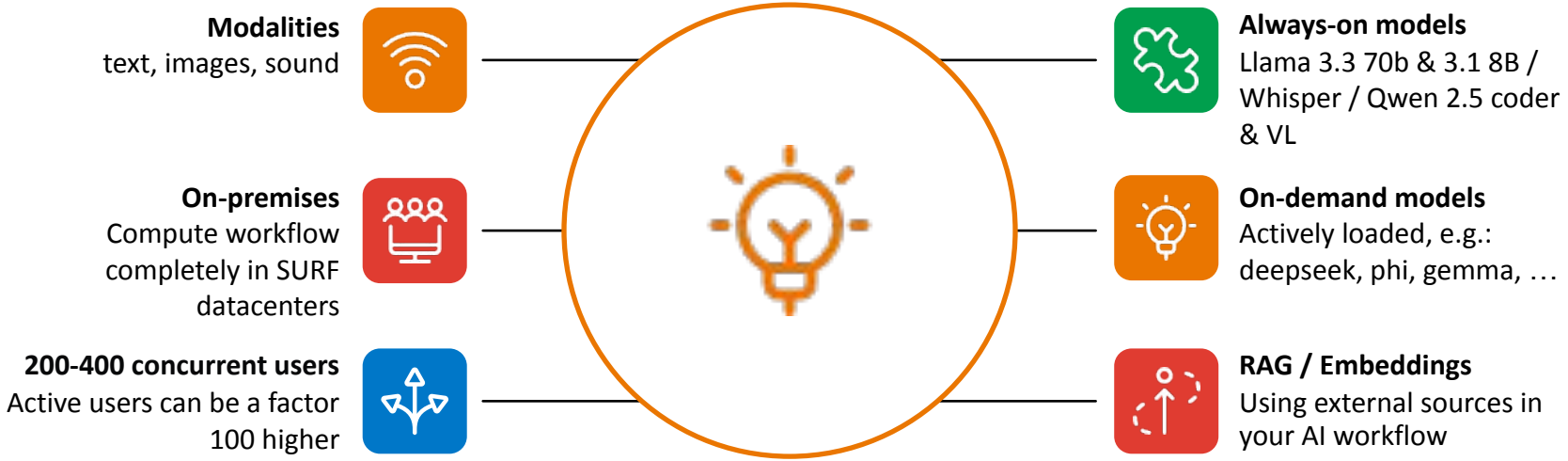
Image by Vectorportal.com

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AI Hub: sovereign platform for AI-inference



What can we do in the pilot phase



Simplified summary: where to go?

- For ease of use and plug and play compute: go to Research Cloud.
- For very powerful parallel computing; go to Snellius.
- For extremely large (parallel) computations, go to LUMI.
- For structurally processing very large amounts of data: go to Spider.
- For easy storage and data sharing; go to Research Drive.
- For cheap long-term storage; go to Data Archive.
- For AI and Machine learning support; to our AI expertise center.
- For advice and POC's on public clouds; go to our Cloud Consultancy team.
- For sovereign AI-inference platform; go to the AI Hub.

SURF HPML Group

Bryan Cardenas
September 2025

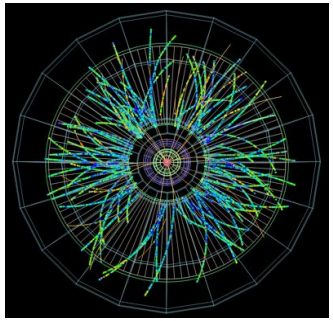
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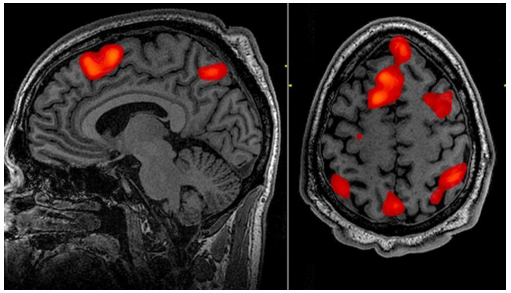
Recipe for Assistance

- Provide Hardware & Storage
- Run Software On Snellius
- Program Efficiency
- Algorithmic Design

HPML Projects



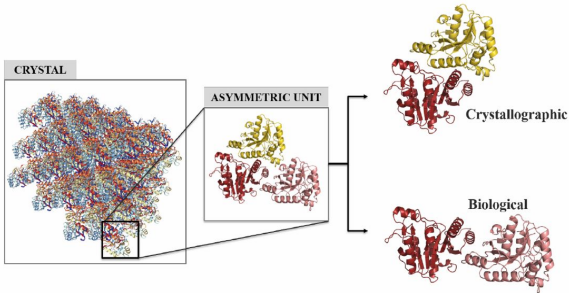
DL models for High Energy Particle Physics **CERN**



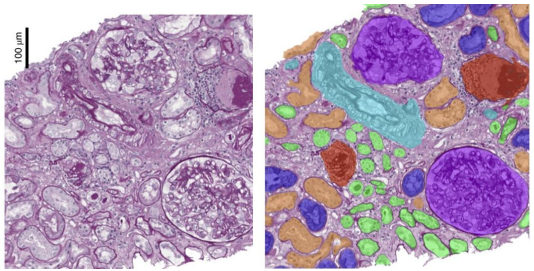
FMRI & EEG **AMC**



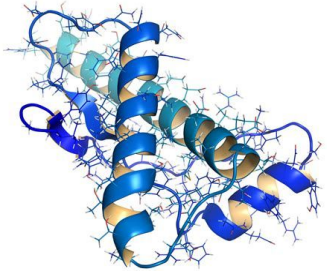
Agents for BioChemistry Research Papers **UMC**



Biomolecular interfaces from crystal artifacts **Universiteit Utrecht**

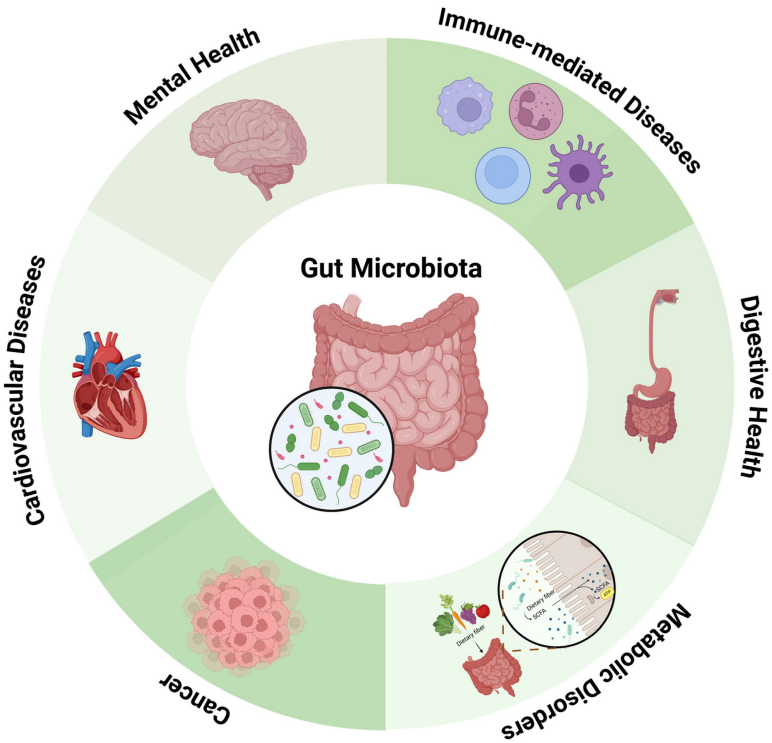


Multimodal analysis and model training for histopathology **Radboud & AMC**



Protein Folding & Design **Intel & AMC**

Gut-Brain Interplay



Hereditary

01. Neurological Diseases

02. Gut Microbiome Disorders

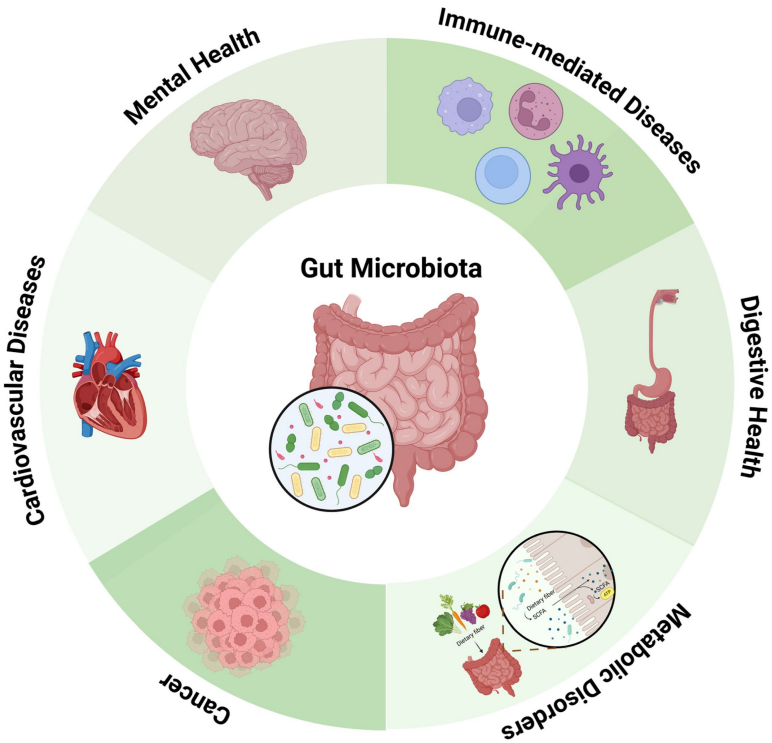
Multi-Modality

Data Assembly

Multi-Linguality

Privacy & Security

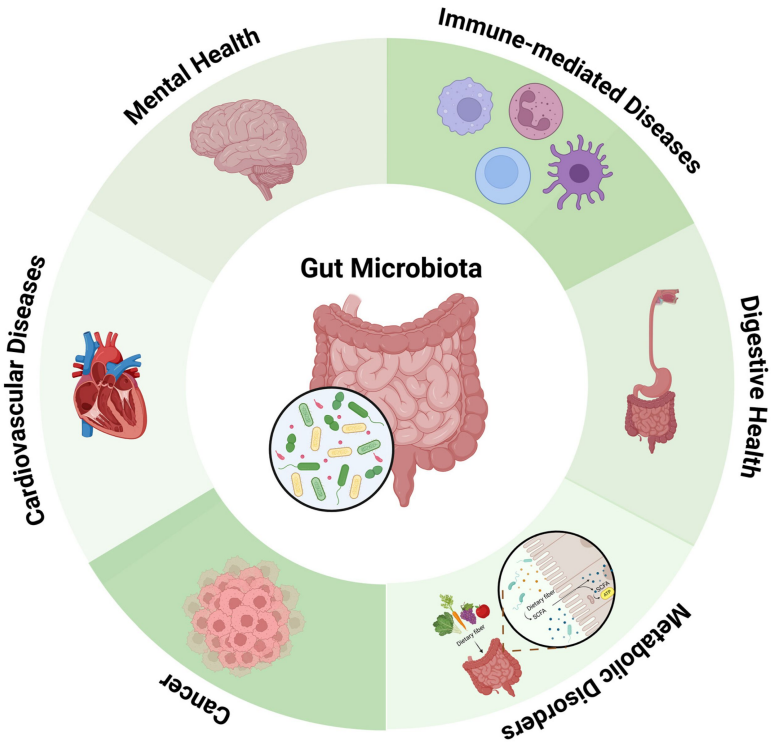
Gut-Brain Interplay



Hereditary Data Types



Gut-Brain Interplay



Federated Learning

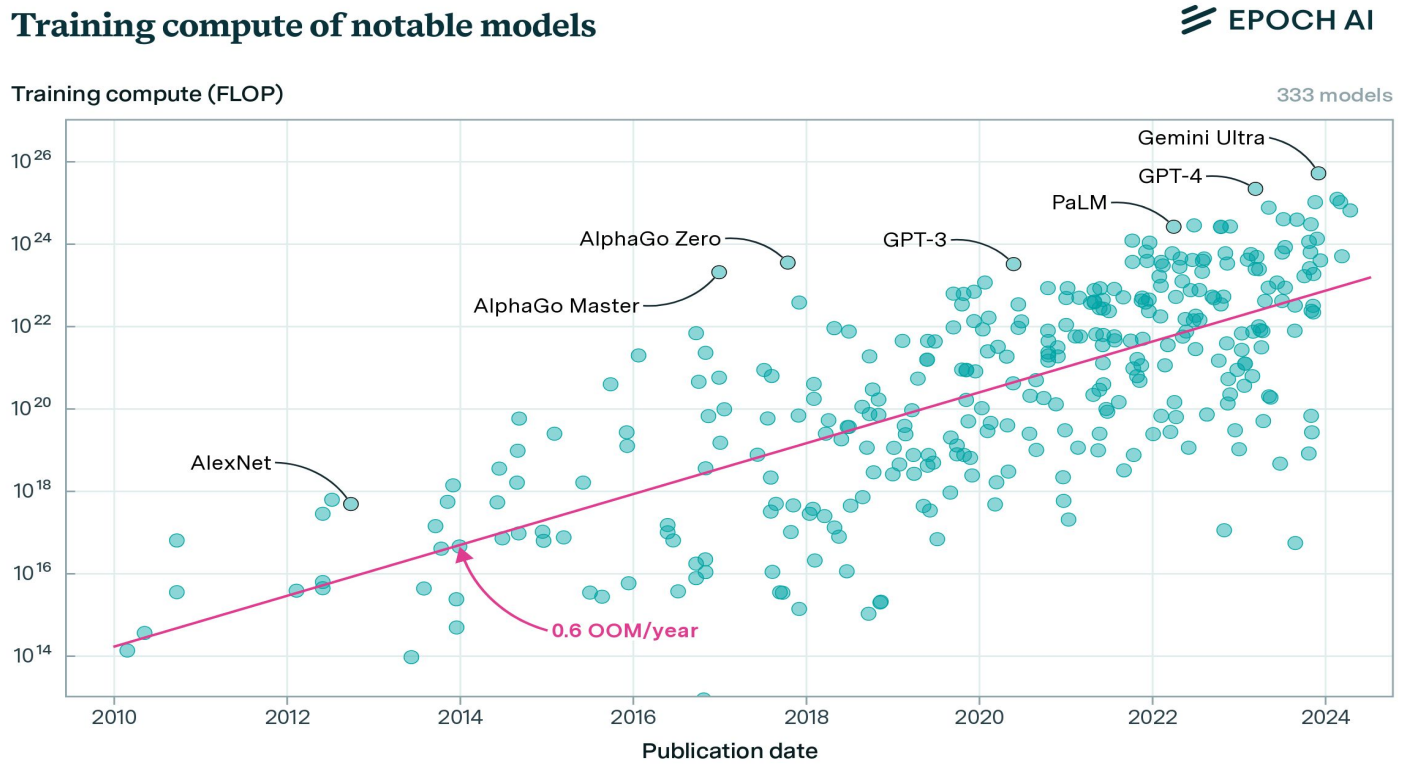
01. Improve diagnostic accuracy

02. Make data comply with privacy and ethics

03. Multiple centers benefit!

SURF

HPML Projects



Q&A